Feature-based Propensities for Counterfactual Learning to Rank

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Abstract

Counterfactual learning-to-rank (LTR) optimizes ranking models from biased implicit click feedback. The prominent policy-aware approach relies on inverse-propensity-scoring (IPS) where observed clicks on documents are reweighted inversely to the probability that users examine them given the logging policy (a.k.a. the propensity). In theory, these propensities are a simple expectation over display positions; but in practice, the placement probabilities are often unknown and instead the observed frequency per position is used. This forms a substantial issue in settings with long-tail query distributions where most query-document combinations are only observed once in a dataset, and therefore, no reliable propensity estimate can be made. Consequently, this could negate the bias correction of IPS-based methods.

To address this issue, we propose feature-based propensity estimation, where propensities are predicted based on query-document features. Thereby, we aim to provide stable propensity estimates for all query-document pairs regardless of the number of observations that are available for a pair. We theoretically show that feature-based propensity estimation can be performed if the logging policy behavior is based on query-document features. Experiments in semi-synthetic data show that feature-based propensities can recover logging policy propensities more effectively than frequency-based baselines and significantly improve performance of counterfactual LTR methods in diverse settings, with especially large improvements in sparse-data settings. All code, data, and results are available at: https://github.com/stanfris/feature-based-propensities-for-counterfactual-ltr.

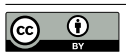
CCS Concepts

• **Information systems** → **Learning to rank**.

Keywords

Counterfactual Learning to Rank, Propensity Estimation, Bias

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1 Introduction

Counterfactual learning-to-rank (LTR) concerns the optimization of ranking models from implicit click feedback for search and recommendation systems [11, 15, 16, 39]. Clicks are a biased signal of user preference as they are also affected by many other factors [1, 28, 36, 37]. Most prominently, users are more likely to examine and thus interact with items that are displayed at the top of rankings, a phenomenon known as *position bias* [7, 9, 14, 29]. Thus, in order to accurately infer relevance from clicks, one needs to correct for such biases [1, 4, 28, 36, 37, 40].

The foundational inverse-propensity-scoring (IPS) estimator reweights clicks to correct for selection bias caused by user examination [16, 39]. The relevance estimates of IPS are proven to be unbiased under the assumptions that the examination model accurately describes user behavior and that the true propensity values are known. Subsequent work has introduced methods for other biases, e.g., the affine method for trust bias [37]; on position bias estimation, e.g., by intervention harvesting [3]; and more reliable estimation, e.g., with doubly-robust (DR) estimation [26].

Most contemporary counterfactual LTR methods rely on policy-aware propensities [27] that take into account both the users' position bias and the stochastic behavior of the logging policy or changes in policy during logging [11, 26, 28]. In their theory, previous work assumes the logging policy is known exactly and does not have to be estimated [26–28]. But this is often not the case in practice; real-world search and recommendation systems have complex behavior that is subject to business rules, which is not directly sampled from a known probability distribution. Moreover, interaction datasets often do not store enough information, and the exact logging policy cannot be retrieved accurately. Thus, besides the estimation of position bias, *estimating logging policy behavior* is also a practical challenge that needs to be overcome for IPS.

Previous studies have used the observed frequency of each position as an estimate of the logging policy [22, 26], although their primary focus was not on reconstructing the logging policy itself. At first glance, this provides a simple and unbiased estimation method; however, two issues arise: Firstly, an unbiased estimate of a propensity can still lead to a biased reciprocal propensity estimate (as we show in Section 4). Secondly, in practice, query distributions are often extremely long tailed and thus very few observations are available per query-document pair. Figure 1 illustrates this problem for two real-world click datasets [18, 42] where the large majority

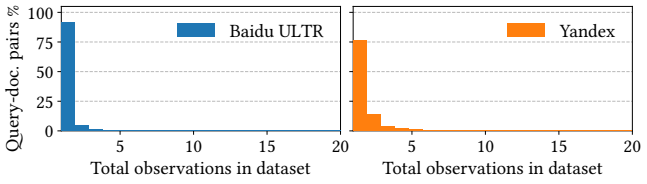


Figure 1: Distribution of observed frequencies of query-document pairs in two real-world web-search click datasets.

of query-document pairs are only observed once. This problem did not arise for previous work that relied on semi-synthetic experimental setups where query distributions were artificially not sparse [26–28]. But in more realistic long-tail settings, propensity estimation cannot be done accurately with observed frequencies, and therefore, truly unbiased IPS seems infeasible in practice.

In this work, we directly address the logging policy estimation problem in long-tail query distribution settings by proposing to estimate propensity values based on query-document features, instead of using observed frequencies directly. We theoretically show that observed-frequency propensities lead to biased reciprocal propensity estimates and that under a mild logging policy assumption, feature-based propensities provide a reduction to this bias. Our experimental results show that the proposed feature-based propensity estimation can accurately recover true propensities and result in better IPS optimization when observed data is sparse, and also in some settings where data is not sparse. Thereby, the contributions of this work increase the robustness and applicability of counterfactual LTR methodology.

2 Related Work

Previous work in counterfactual LTR has addressed several forms of bias and accordingly different propensity estimation methods. Position bias is the most traditional form of bias addressed and can be estimated using EM optimization [4], intervention harvesting [3, 10], or direct randomization [40]. Trust bias occurs when the display position makes users perceive items as more relevant. Addressing it requires estimating an additional parameter per position [2] and an affine variation of IPS [37]. Item-selection bias occurs when not all items can be displayed at once [27, 29] and can be addressed with policy-aware propensities, which require logging policy estimation [27]. Policy-aware propensities can also account for changes in logging policy during the gathering of data [28]. However, estimating the logging policy is often a problem in real-world settings due to changing systems, business rules and a lack of logging contextual information, which make it impossible to directly infer the logging policy. This is exemplified by the many publicly available industry datasets that do not indicate their logging policy [18, 19, 21, 30, 38, 42]. Simultaneously, none of the previous works that have relied on policy-aware propensities have focused on their estimation nor discussed any potential problems with the required logging policy estimation [26–28].

Estimating logging policies is also a relevant challenge for off-policy estimation in the broader field of reinforcement learning [6, 20, 23, 35, 41]. Raghu et al. [32] demonstrated that large errors arise when logging policy behavior is not estimated correctly, and applied an effective solution based on simple clustering techniques for policy estimation. Kato et al. [17] and Narita et al. [24] found that propensities based on estimated logging policies can provide higher stability over using the true logging policy. Zhang et al. [41] use uncertainty-aware reweighting for propensities, based on observed frequency of actions, leading to increased robustness. In the setting of recommender systems, Saito and Joachims [34] propose the marginalized IPS estimator that bases propensities on learned action embeddings, this allows them to better handle the sparsity in observations that results from the enormous action space in that setting. We take inspiration from these works but uniquely focus on the counterfactual LTR setting.

3 Preliminaries and Background

We follow the most common click model from the counterfactual LTR field, the position-based model [7]: Let k indicate a position (a.k.a. rank) and d a document, we assume that the probability of a user clicking at d when displayed at k is a product of a factor $\alpha_k \in [0, 1]$ and the probability that the user finds d relevant:

$$P(C = 1 \mid d, k) = \alpha_k P(R = 1 \mid d) = \alpha_k R_d. \tag{1}$$

Commonly, α_k is interpreted as the probability of the user examining position k, and thus, that this click probability is the joint probability of the user examining k and finding d relevant [7, 33, 37, 40].

At the core of counterfactual LTR is the problem of estimating how many clicks a ranking will receive based on clicks for other rankings. More broadly, we aim to estimate (and subsequently optimize) the expected number of clicks a policy π receives, where π is a probability distribution over rankings $y = [y_1, y_2, \ldots, y_K]$; thus under our assumed click model, our estimand $\mathcal{R}(\pi)$ is:

$$\mathcal{R}(y) = \sum_{k=1}^{K} \alpha_k R_{y_k}, \quad \mathcal{R}(\pi) = \mathbb{E}_{y\sim\pi}[\mathcal{R}(y)] = \sum_{d\in D} \omega_d R_d, \tag{2}$$

where $\omega_d = \mathbb{E}_{y\sim\pi}[\alpha_{k(d)}] = \sum_{k=1}^{K} \pi(k \mid d)\alpha_k$ with $\pi(k \mid d)$ denoting the probability that d is placed at k by π and $D = \{d_1, d_2, \ldots\}$ the set of available documents. For brevity, we follow previous work and only consider a single query for our mathematical notation, this can be easily extended to multiple queries [16, 26, 28].

Formally, our goal becomes to unbiasedly estimate $\mathcal{R}(y)$ based on a set of N i.i.d. interactions gathered on rankings by the logging policy π_0: let $c_i(d) \in \{0, 1\}$ denote whether d was clicked in interaction i. The main issue here is that each click probability is a product of α_k and R_d, thus in order to be unbiased, we need to correct for the influence of α on c. All of the estimators discussed in this work make use of policy-aware propensities [27], these reflect the expected influence of α on the click probabilities under π_0:

$$\rho_d = \mathbb{E}_{y\sim\pi_0}[\alpha_{k(d)}] = \sum_{k=1}^{K} \pi_0(k \mid d)\alpha_k, \quad \hat{\rho}_d \approx \rho_d. \tag{3}$$

Because α cannot be observed directly, an estimate of ρ has to be used, which we denote by $\hat{\rho}$. We note that ω and ρ are very similar but different: ω concerns expected values under π, the policy we are evaluating; whilst ρ concerns the logging policy π_0. Analogously, $\hat{\omega}_d \approx \omega_d$ is also needed. We can now discuss the estimators.

Inverse-propensity-scoring (IPS). The IPS estimator with policy-aware propensities [27] is an extension of earlier IPS estimators [16, 39]. To correct for position bias in observed clicks, it weights each click by the estimated ratio between the corresponding ω_d and ρ_d:

$$\hat{\mathcal{R}}_{\text{IPS}}(\pi) = \frac{1}{N} \sum_{i=1}^{N} \sum_{d\in D} \frac{\hat{\omega}_d}{\hat{\rho}_d} \Big(c_i(d)\Big). \tag{4}$$

It is unbiased if the ratios are correctly estimated [26, 27]:

$$(\forall d \in D,\ \rho_d = \hat{\rho}_d \wedge \omega_d = \hat{\omega}_d) \longrightarrow \mathbb{E}[\hat{R}_{\text{IPS}}] = R(\pi). \tag{5}$$

The downside of IPS estimation is that large values for the ratios $\hat{\omega}_d / \hat{\rho}_d$ (usually due to small $\hat{\rho}_d$ values) result in enormous variance.

The Direct-method (DM). The DM proposes a low-variance alternative to IPS that completely relies on regression estimates of relevance $\hat{R}_d \approx R_d$ instead of reweighting clicks [8, 26]:

$$\widehat{\mathcal{R}}_{DM}(\pi) = \sum_{d \in D} \hat{\omega}_d \hat{R}_d. \tag{6}$$

By completely ignoring observed clicks it avoids high variance but requires all regression estimates to be correct for unbiasedness:

$$(\forall d \in D,\ R_d = \hat{R}_d \wedge \omega_d = \hat{\omega}_d) \longrightarrow \mathbb{E}[\hat{R}_{\text{DM}}] = R(\pi). \tag{7}$$

This is a practically infeasible objective, as finding accurate regression estimates is as difficult as the core counterfactual LTR problem itself. It can be useful when variance is a greater problem than bias.

Doubly-robust (DR) Estimation. DR estimation starts with the DM estimate and then corrects its bias with IPS, thereby it is able to combine desirable aspects of both [8, 26]. It does this by estimating the error of DM with IPS and subtracting it from the DM estimate:

$$\hat{\mathcal{R}}_{DR}(\pi) = \hat{\mathcal{R}}_{DM}(\pi) + \frac{1}{N} \sum_{i=1}^{N} \sum_{d \in D} \frac{\hat{\omega}_d}{\hat{\rho}_d} \Big(c_i(d) - \hat{\alpha}_{k_i(d)} \hat{R}_d \Big), \tag{8}$$

where $k_i(d)$ denotes the display position of d at interaction i. As we can see, it reweights the difference between the observed click $c_i(d)$ and the predicted click probability, $\hat{\alpha}_{k_i(d)} \hat{R}_d$ to estimate the error of DM [26]. It has more favorable unbiasedness conditions and generally a lower variance than IPS, but these depend on how $\hat{\rho}$ and $\hat{\omega}$ are estimated, we refer to [26] for a further discussion.

4 Frequency-based Propensity Estimation

The estimators in Section 3 have all made use of the (policy-aware) propensity estimates $\hat{\rho}_d$ which are estimates of the true propensity ρ_d. From their definitions in Eq. 3, we see that there are two parts to estimating ρ_d: estimating the position bias variables α_k and estimating the logging policy distribution $\pi_0(k \mid d)$. Much previous work has been devoted to estimating position bias [3, 4, 10, 40]. For this work, we assume that estimates $\hat{\alpha}_k \approx \alpha_k$ are available, and focus on estimating the policy behavior.

The most straightforward method for estimating the logging policy behavior is by using the observed frequency, we call this the *frequency-based* propensity estimator. It simply takes the observed placements of d to compute the mean α_k over those positions:

$$\tilde{\rho}_d := \frac{1}{N} \sum_{i=1}^{N} \hat{\alpha}_{k_i(d)}. \tag{9}$$

Due to its simplicity, it is perhaps not a surprise that this estimator has previously been used in existing work [22, 26].

Given that interactions are i.i.d., the expected value of the frequency-based propensity estimator for a document d can be written as:

$$\mathbb{E}_{k \sim \pi_0}[\tilde{\rho}_d] = \mathbb{E}_{k \sim \pi_0}\left[\hat{\alpha}_{k(d)}\right] = \sum_{k=1}^{K} \pi_0(k \mid d) \hat{\alpha}_k. \tag{10}$$

Comparing Eq. 3 with Eq. 10, we can see that this estimator is unbiased with respect to ρ if $\hat{\alpha}_k$ is correctly estimated for all ranks:

$$(\forall k,\ \hat{\alpha}_k = \alpha_k) \longrightarrow \bar{\rho}_d = \rho_d. \tag{11}$$

However, the unbiasedness of the frequency-based propensity estimate $\tilde{\rho}_d$ does not guarantee that an IPS estimator using it will be unbiased. This is because IPS uses the reciprocal of the propensity, and by Jensen's Inequality, since the function $f(x) = 1/x$ is convex:

$$\frac{1}{\mathbb{E}_{k \sim \pi_0}[\tilde{\rho}_d]} = \frac{1}{\rho_d} \leq \mathbb{E}_{k \sim \pi_0}\left[\frac{1}{\tilde{\rho}_d}\right]. \tag{12}$$

Therefore, applying the reciprocal propensity estimate $1/\tilde{\rho}_d$ may introduce an upward bias in the IPS weight.

We can approximate this bias by applying the Taylor expansion for the first moment:

$$\mathbb{E}[f(X)] \approx f(\mu_X) + \frac{f''(\mu_X)}{2} \sigma_X^2.$$

Here, f is the applied approximation function with respect to random variable X, where μ_X, σ_X are the mean and standard deviation respectively. In our case $X = \tilde{\rho}_d$ and $f(x) = 1/X$. Thus, we also have the derivatives: $f'(X) = -1/X^2$ and $f''(X) = 2/X^3$. We can now apply these in the Taylor expansion to get the expected value:

$$\mathbb{E}_{k \sim \pi_0}\left[\frac{1}{\tilde{\rho}_d}\right] \approx \frac{1}{\mathbb{E}_{k \sim \pi_0}\left[\hat{\alpha}_{k(d)}\right]} + \frac{\mathbb{V}_{k \sim \pi_0}(\tilde{\rho}_d)}{\mathbb{E}_{k \sim \pi_0}\left[\hat{\alpha}_{k(d)}\right]^3}. \tag{13}$$

We can then consider the variance of the estimate $\tilde{\rho}_d$:

$$\mathbb{V}_{k \sim \pi_0}(\tilde{\rho}_d) = \frac{1}{N^2} \sum_{i=1}^{N} \mathbb{V}_{k \sim \pi_0}(\hat{\alpha}_{k_i(d)}) = \frac{1}{N} \mathbb{V}_{k \sim \pi_0}(\hat{\alpha}_{k_i(d)}), \tag{14}$$

where we use the assumption that samples from the logging policy are i.i.d. across interactions. We can combine this with Eq. 13 to get an approximation of the expected value in the frequency-based reciprocal propensity estimate:

$$\mathbb{E}_{k \sim \pi_0}\left[\frac{1}{\tilde{\rho}_d}\right] \approx \frac{1}{\mathbb{E}_{k \sim \pi_0}\left[\hat{\alpha}_{k(d)}\right]} + \frac{\mathbb{V}_{k \sim \pi_0}(\hat{\alpha}_{k_i(d)})}{N\, \mathbb{E}_{k \sim \pi_0}\left[\hat{\alpha}_{k(d)}\right]^3}. \tag{15}$$

Clearly, when the correct position bias is given ($\forall k,\ \alpha_k = \hat{\alpha}_k$), its bias is (approximately) $\mathbb{V}_{k \sim \pi_0}(\hat{\alpha}_{k_i(d)}) / (N\, \mathbb{E}_{k \sim \pi_0}\left[\hat{\alpha}_{k(d)}\right]^3)$. Therefore, this bias will be greater for small expected propensities and large variance in the α_k assigned to a document. Additionally, we see that bias diminishes as N becomes larger, and vice versa, increases for smaller N values. This shows that whilst frequency-based policy-aware propensity estimates are unbiased, they cannot be used directly to obtain an unbiased estimate of reciprocal propensities. Moreover, our theoretical analysis shows that this problem is exacerbated in long-tail settings where the number of observations (N) is very low for the majority of query-document pairs.

5 Method: Feature-Based Propensity Estimation

In this section, we present our novel *feature-based* propensity estimation method which predicts policy-aware propensity values based on query-document features.

5.1 Defining feature-based propensities

In Section 4 we discussed frequency-based propensity estimation and showed that it does not provide unbiased estimates of the reciprocal (policy-aware) propensities, which are used in IPS estimation. In particular, the bias of an estimated reciprocal propensity is inversely tied to the number of observations for the corresponding query-document pair. The aim of this work is to find an alternative estimation method that does not rely on observed frequencies, and thereby, avoids requiring large numbers of observations which

are not available in long-tail settings. For our approach, we take inspiration from the marginalized IPS estimator from Saito and Joachims [34] which bases propensities on action embeddings that can represent categories or clusters of items, in order to deal with large action spaces.

In our setting, we assume there is a feature vector representation x_d available for each query-document pair d in the set D. Furthermore, we also assume that the set D is a set of samples from a larger distribution of query-document pairs denoted by $\mathcal{D}$. Formally, we define a feature-based propensity for a feature vector x to be the expected value of the policy-aware propensity of documents with identical feature vectors:

$$\rho_x := \mathbb{E}_{d\sim\mathcal{D}}[\rho_d \mid x_d = x]. \tag{16}$$

Correspondingly, the feature-based propensity for a document d:

$$\rho_{x_d} := \mathbb{E}_{d'\sim\mathcal{D}}[\rho_{d'} \mid x_{d'} = x_d]. \tag{17}$$

Thus, by basing the propensities on features instead of identities, propensities can be shared with documents with identical features. This reduces the need for many observations per query-document pair, as the propensities of once-observed pairs can be based on observations of other pairs as well. In the remainder of this section, we theoretically analyze when our propensities lead to unbiased IPS estimates, and propose two methods for estimating their values in practice.

5.2 Conditions for unbiased IPS estimation

The purpose of the feature-based propensities is to provide unbiased IPS estimation without requiring numerous observations per query-document pair. We are able to show that under certain conditions feature-based propensities are equivalent to standard policy-aware propensities, and thus, they provide the same guarantees in those conditions. In particular, if we assume that logging policy behavior is only determined by document features:

$$\forall(k,d) \in K \times D, \quad \pi_0(k \mid d) = \pi_0(k \mid x_d), \tag{18}$$

then using Eqs. 3 and 17, we have that for any document d:

$$\begin{aligned}\rho_{x_d} &= \mathbb{E}_{k\sim\pi_0}[\rho_d \mid x_d] = \mathbb{E}_{k\sim\pi_0}\left[\sum_{k=1}^{K}\pi_0(k \mid d)\alpha_k \,\middle|\, x_d\right] \\ &= \sum_{k=1}^{K}\mathbb{E}_{k\sim\pi_0}[\pi_0(k \mid d) \mid x_d]\alpha_k = \sum_{k=1}^{K}\pi_0(k \mid x_d)\alpha_k = \rho_d.\end{aligned} \tag{19}$$

In other words, if the features in x_d contain all the information that π_0 used to determine its behavior then the original policy-aware propensity ρ_d for a document d is equal to its corresponding feature-based propensity ρ_{x_d}. Consequently, in this scenario, any IPS or DR estimators using either of the two types of propensities have the same properties.

We note that the requirement for x_d to contain all information that the logging policy used appears quite feasible in practice, but has a few caveats. Firstly, it requires that this information is logged in completion, which is not always the case in public datasets where the information can be sensitive, or in scenarios where some features are dynamic and their values change continuously over time but are not logged instantly. Secondly, for ranking models, the decision on where to place a document d also depends on where all other documents d' are placed, thus x_d could have to contain (aggregate) information about the entire candidate set. In turn, this can make the chance that documents share identical features very small; in response to this issue, we propose two pragmatic solutions based on clustering and supervised learning in Section 5.4.

5.3 Bias of reciprocal propensity estimation

Our introduction of feature-based propensities was motivated by the bias of the frequency-based estimates of reciprocal policy-aware propensities. To illustrate how our feature-based propensities can provide an improvement, we analyze the bias of the reciprocal propensity when using a feature-based estimate.

For this analysis, we need an estimate $\hat{\rho}_{x_d} \approx \rho_{x_d}$, we also use observed frequencies but instead of using those of individual query-document pairs, we use them for a group of query-document pairs with the same features. Formally, we define a grouping function $G(d)$ that represents all observed documents with features identical to d, let $\mathbb{D}$ be all our observed data, then $G(d) := G_d := \{d' \in \mathbb{D} : x'_d = x_d\}$, where G_d is introduced for brevity. Importantly, whilst our notation thus far has only considered a single query at a time, the data in $\mathbb{D}$ can cover multiple queries, therefore, we introduce N_d to denote the number of observations of query-document pair in $\mathbb{D}$. Our feature-based propensity estimate is then:

$$N_{G_d} := \sum_{d'\in G_d} N_{d'}, \quad \hat{\rho}_{G(x_d)} := \frac{1}{N_{G_d}}\sum_{d'\in G_d}\sum_{k_i\in\mathbb{D}\,:\,d'\in k_i}\hat{\alpha}_{k_i(d')}. \tag{20}$$

In other words, $\hat{\rho}_{G(x_d)}$ is the observed mean $\hat{\alpha}_k$ value observed for query-document pairs with features identical to x_d. Importantly, the number of observations is strictly equal or greater than for that of any individual query-document pair: $N_{G_d} \geq N_d$.

The following analysis is analogous to that of Section 4 but now applied to $\hat{\rho}_{G(x_d)}$ instead of the frequency-based propensity $\tilde{\rho}$. An added complexity is that the G_d set is now also a random variable to consider; for simplicity and brevity we condition on N_{G_d}:

$$\mathbb{E}_{G_d\sim\mathcal{D},k\sim\pi_0}\left[\hat{\rho}_{G(x_d)} \mid N_{G_d}\right] = \mathbb{E}_{d'\sim\mathcal{D},k\sim\pi_0}\left[\hat{\alpha}_{k(d')}\right], \tag{21}$$

since observations are assumed i.i.d., the variance of $\hat{\rho}_{G(x_d)}$ is:

$$\mathbb{V}_{k\sim\pi_0}(\hat{\rho}_{G(x_d)}|\,N_{G_d}) = \frac{1}{N_{G_d}}\mathbb{E}_{d'\sim\mathcal{D}}\left[\mathbb{V}_{k\sim\pi_0}(\hat{\alpha}_{k(d')})|\,x_{d'} = x_d\right]. \tag{22}$$

Applying the same Taylor expansion for the first moment as laid out in Section 4 results in the following approximation for the expected value of the reciprocal propensity estimate:

$$\begin{aligned}&\mathbb{E}_{G_d\sim\mathcal{D},k\sim\pi_0}\left[\frac{1}{\hat{\rho}_{G(x_d)}}\,\middle|\,N_{G_d}\right] \\ &\quad\approx \frac{1}{\mathbb{E}_{d'\sim\mathcal{D},k\sim\pi_0}\left[\hat{\alpha}_{k(d')}\right]} + \frac{\mathbb{E}_{d'\sim\mathcal{D}}\left[\mathbb{V}_{k\sim\pi_0}(\hat{\alpha}_{k(d')})|\,x_{d'} = x_d\right]}{N_{G_d}\,\mathbb{E}_{d'\sim\mathcal{D},k\sim\pi_0}\left[\hat{\alpha}_{k(d')}\right]^3}.\end{aligned} \tag{23}$$

It is difficult to compare this result with Eq. 15 due to the additional random variable. However, if we assume that logging policy behavior is only determined by document features (c.f. Eq. 18 and Section 5.2) then the result can be simplified. To start, the expected reciprocal propensity becomes:

$$\begin{aligned}&(\forall(k,d) \in K \times D,\ \pi_0(k \mid d) = \pi_0(k \mid x_d)) \\ &\qquad\longrightarrow \left(\mathbb{E}_{G_d\sim\mathcal{D},k\sim\pi_0}\left[\hat{\rho}_{G(x_d)} \mid N_{G_d}\right] = \mathbb{E}_{k\sim\pi_0}\left[\hat{\alpha}_{k(d)}\right]\right).\end{aligned} \tag{24}$$

The variance of the expected propensity becomes:

$$(\forall (k,d) \in K \times D,\ \pi_0(k \mid d) = \pi_0(k \mid x_d)) \longrightarrow \left(\mathbb{V}_{k\sim\pi_0}(\hat{\rho}_{G(x_d)} \mid N_{G_d}) = \frac{1}{N_{G_d}} \mathbb{V}_{k\sim\pi_0}(\hat{\alpha}_{k(d)}) \right). \quad (25)$$

The approximation of the expected reciprocal propensity becomes:

$$(\forall (k,d) \in K \times D,\ \pi_0(k \mid d) = \pi_0(k \mid x_d)) \longrightarrow \left(\mathbb{E}\left[\frac{1}{\rho_{G(x_d)}} \,\middle|\, N_{G_d} \right] \approx \frac{1}{\mathbb{E}_{k\sim\pi_0}\left[\hat{\alpha}_{k(d)}\right]} + \frac{\mathbb{V}_{k\sim\pi_0}(\hat{\alpha}_{k(d)})}{N_{G_d}\, \mathbb{E}_{k\sim\pi_0}\left[\hat{\alpha}_{k(d)}\right]^3} \right). \quad (26)$$

This result is directly comparable with Eq. 15; because $N_{G_d} \geq N_d$ (N_d is denoted as N in Eq. 15) the bias of the feature-based estimate guaranteed to be less or equal than that of the frequency-based estimate of Section 4, under the condition that the logging policy behavior is fully determined by the features in x_d. We see that this difference stems directly from the fact that the feature-based estimate (Eq. 20) is based on more observations than the frequency-based estimator (Eq. 9), and thus, it has less bias when $N_{G_d} > N_d$ (under the aforementioned logging policy assumption). Whilst these are real benefits, our estimate is impractical in its requirement of needing to observe query-document pairs with identical features. The remainder of this section proposes more practical implementations of feature-based propensity estimation.

5.4 Practical estimation and implementation

We have defined feature-based propensities, discussed the conditions under which they provide unbiased IPS estimation and analyzed the bias of a simple but naive estimate of reciprocal feature-based propensities. We consider the simple estimate (Eq. 20) naive because it assumes a set of query-document pairs with identical features is available. In practice this is rather unlikely, to address this issue, we propose two estimation methods with more realistic requirements based on clustering and regression respectively.

Clustering-based feature-based propensities. First, we propose using clustering methods to create groups that can be used instead of G_d in Eq. 20. A simple clustering approach is to consider all d' with a Euclidean distance to d below a threshold ε as part of the set: $G_\varepsilon(x_d) = \{d' \in \mathbb{D} \mid \|x_d - x_{d'}\|_2 \leq \varepsilon\}$. This relies on the assumption that a query-document pair d' would be treated identical to d by the logging policy if their features are within an ε distance of each other. In practice, this requires tuning the threshold ε: setting it too large can introduce bias, while setting it too low may prevent enough query-document pairs from falling within the threshold to achieve a substantial variance reduction.

Regression-based feature-based propensities. Second, we propose using a fitted regression model to predict the values of feature-based propensities given their feature values. For this we define $f_\theta(x_d)$ as a regressor that estimates feature-based propensities for feature vector x_d using the parameters θ. Here f_θ can be any regression model that can be optimized with supervised machine learning. In our experiments, we use a feed-forward neural network. For the optimization of the parameters θ, we propose minimizing the mean squared error (MSE) with the observed $\hat{\alpha}$ values:

$$\mathcal{L}(\theta) = \frac{1}{N\,D} \sum_{i=1}^{N} \sum_{d=1}^{D} \left(f_\theta(x_d) - \hat{\alpha}_{k_i(d)} \right)^2. \quad (27)$$

The resulting f_θ can then predict feature-based propensities for any query-document pair, even if they have been previously unseen. This alleviates the dependency on having many observations for accurate estimates, however, it does rely on the regressor being able to accurately capture the relation between features and the true feature-based propensities.

The remainder of this paper experimentally investigates our proposed estimation approaches in semi-synthetic search settings.

6 Experimental Setup

The experiments of this work aim to answer two research questions:

RQ1: Can feature-based propensity estimation effectively recover the true policy-aware propensities?

RQ2: Do feature-based propensities lead to improved performance of counterfactual learning-to-rank methods?

In other words, we evaluate the effectiveness of our proposed feature-based propensity estimation method in terms of its ability to estimate the true propensity values and the downstream performance when using them for counterfactual LTR.

We perform our experiments in a semi-synthetic setup based on three LTR datasets with expert-annotated relevance labels between 0-4. Specifically, we use the MSLR-WEB30K [30] dataset from the Bing search engine (31,531 queries and 136-dimensional feature vectors), the Istella-S [21] dataset (33,018 queries and 220-dimensional feature vectors), and the Yahoo! Webscope [5] dataset (both sets, 29,921 queries and 699-dimensional feature vectors). We apply the feature-transformation $\log1\mathrm{p}(x) = ln(1 + |x|) \odot \mathrm{sign}(x)$ following previous work [13, 31], and conduct top-$K = 25$ ranking settings.

Our logging policy ranking model is a feed-forward neural network (2 layers, 32 neurons, and ELU activations) trained on 10% of the data with ground-truth labels using a pointwise MSE loss; this results in a ranking model that has adequate performance but still leaves sufficient room for improvement. On top of this model, we use the epsilon-greedy policy from Hager et al. [13] for our logging policy: with probability epsilon (ϵ), documents are ranked uniformly at random; otherwise, they are ranked deterministically by logging-policy scores. Unless stated otherwise, $\epsilon = 0.5$.

A benefit of our logging policy is that the true value of the policy-aware propensities of each query-document pair (ρ_d) can be computed exactly, we refer to these as the oracle propensities.

We simulate user clicks following Eq. 1. Our position bias values are $\alpha_k := 1/k$, where k is the rank of the query-document pair [13]. At each click interaction, a query is sampled uniformly over the training and validation partitions and a corresponding ranking is sampled from the logging policy [26]. Relevance probabilities R are made by mapping the relevance labels in the dataset ($\mathrm{label}_d \in \{0, 1, 2, 3, 4\}$) by $R_d := \frac{1}{1-\exp(\mathrm{label}_d+2)}$, following the setup of Hager et al. [13]. We simulate a varying number of clicks, but always over the same query-document dataset, for a fair comparison between sample sizes. Clicks are resampled for every run. Unless stated otherwise, estimators are provided the position bias values $\hat{\alpha}_k := \alpha_k$.

Our optimization of the ranking models for the new policies is adapted from Oosterhuis [26]. We apply feed-forward neural networks (2 layers, 32 neurons, and ELU activations) as Plackett-Luce ranking models optimized with policy gradients estimated with PL-Rank-2 [25]. We train for up to 100 epochs, with early stopping

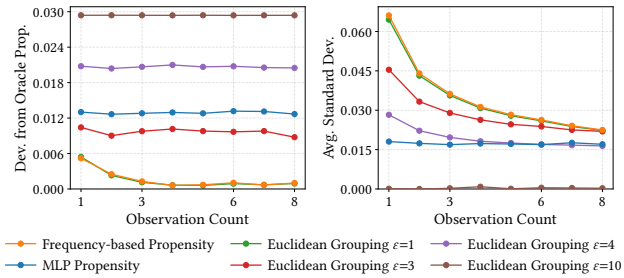


Figure 2: Mean and standard deviation of propensities estimated using frequency-based, clustering-based methods and supervised-learning-based MLP methods ($N = 30000$).

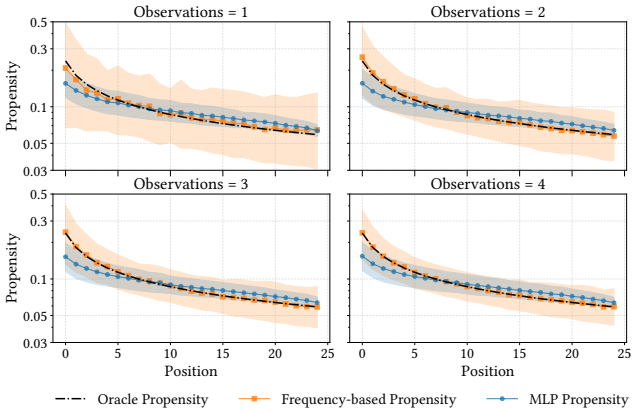


Figure 3: Frequency-based propensity estimation compared to propensities estimated using a feature-based MLP for different numbers of observations and per mode-position of documents. Shaded areas show standard deviations.

after three epochs of no improvement in validation loss. For the regression estimates of DM and DR, we apply the model and bias corrected cross-entropy loss from Oosterhuis [26]. As this loss also reweighs using inverse propensities, we can apply feature-based propensities here as well. Our primary metric is the normalized discounted cumulative gain (NDCG) metric. Results for experiments that measure performance are based on 20 independent runs and report statistical significance based on a two-sided Student's t-test.

7 Results: Feature-Based Propensity Estimation

In this section, we address **RQ1**: *Can feature-based propensity estimation effectively recover the true policy-aware propensities?* In order to compare our methods to the existing frequency-based propensity methods, we conduct experiments on a subset of MSLR-WEB30K. Following Hager et al. [13], we truncate each query to its 25 most relevant documents. This reduces the number of query samples needed to obtain a distribution that includes frequently observed pairs, while ensuring the logging policy can meaningfully differentiate between ranks. We perform the feature-based clustering method based on Euclidean distance. We also apply the feature-based regression model, tuned using a grid search over the number of layers $\{2, 3, 4\}$, neurons $\{64, 128, 256\}$, and dropout $\{0.0, 0.1, 0.2, 0.3\}$ over $N = 50000$ sessions. The final model has 4 layers, 128 neurons, and no dropout, and will be referred to as the multilayer perceptron (MLP). For evaluation, we group query-document pairs by the number of times they have been observed after sampling $N = 30000$ sessions over the truncated test set, which was chosen as it introduces an even distribution of observations over the queries with sufficient samples up to 8 observations. We compare the effectiveness of the frequency-based propensity estimator, and the feature-based clustering and MLP methods for propensity estimation in Figure 2. Here, the resulting mean absolute error (mean deviations) from the simulated propensity and average standard deviations for between 1 and 8 observations for query-document pairs are shown. We observe the frequency-based propensities have a low mean deviation, which is expected as they are unbiased, but also have high average standard deviations, especially when few observations are present. For the Euclidean clustering methods, we observe that with increasing group thresholds, the deviation from the oracle propensity is increased, while the average standard deviation is reduced compared to the frequency-based estimator. Thus applying larger clustering thresholds leads to increased bias but further reduced variance. This suggests a bias-variance trade-off: less strict clustering reduces variance but increases bias, while stricter clustering limits bias at the cost of smaller variance reduction.

MLP predicted propensities make a tradeoff between absolute error and standard deviation that clustering is unable to achieve, it has a lower or comparable standard deviation than clustering with $\varepsilon = 4$ (depending on the number of observations) and also a substantially lower absolute error. Interestingly, the standard deviation for MLP remains nearly constant as the number of observations changes. This seems to indicate that the MLP model prediction generalizes across documents and does not require a large number of observations for every document-query pair for reliable estimation.

We further examine the effectiveness of feature-based MLP propensity estimation by evaluating estimated propensities across positions in Figure 3. Here, we group documents by their mode-position, their most likely position under the logging policy, and show average propensity estimates and standard deviations for each position, with up to four document observations. For documents with one or two observations, the feature-based estimator has a much lower standard deviation than the frequency-based estimator across all positions, indicating that it predicts placement well even with limited observations. We do observe some bias for the MLP model as its mean is slightly below the true average at low positions and above it at high positions, suggesting a structural bias. This likely results from the MSE loss, which encourages conservative predictions. As the number of observations increases, the variances of the frequency-based and MLP regression-based estimates become more similar, although the latter generally retains a lower variance.

To conclude, we answer **RQ1** in the affirmative: feature-based propensity estimators can effectively recover the simulated logging policy, substantially reducing variance while only marginally increasing bias compared to frequency-based propensity estimates. In particular, we find that using MLP regression recovers propensities effectively regardless of the number of observations, making it more robust than clustering. Because of this conclusion, we only use MLP regression for the feature-based propensity estimator in the following section.

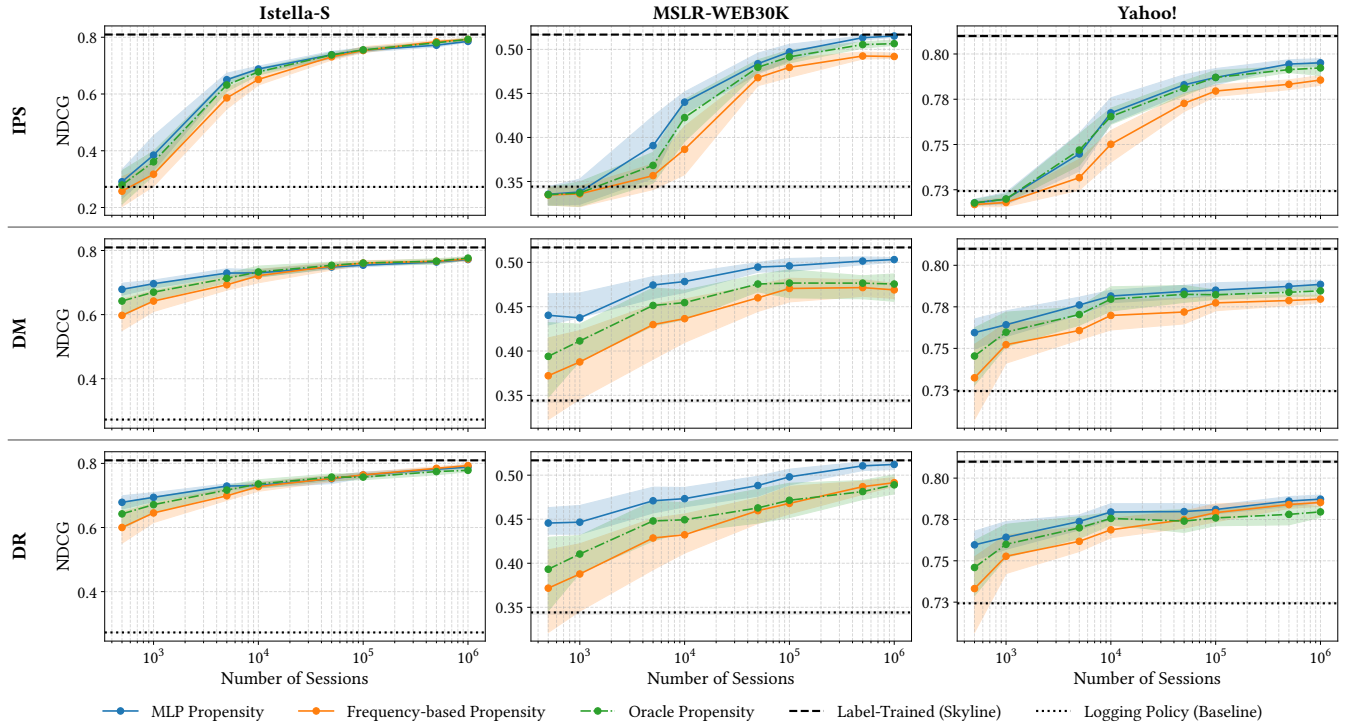


Figure 4: Performance of IPS, DM and DR estimation with different propensity estimation methods on NDCG at different sample sizes. Results are means over 20 independent runs, shaded areas indicate the 90% prediction intervals.

8 Results: Counterfactual LTR with Feature-Based Propensity Estimation

In this section, we investigate how improvements in propensity estimation translate to counterfactual LTR performance on semi-synthetic datasets. We aim to answer **RQ2**: *Do feature-based propensities lead to improved performance of counterfactual learning-to-rank methods?* Our evaluation covers several different settings: We first investigate different sample sizes, subsequently different logging policies, and finally, performance under estimated position bias.

Effect of Sample Size. We first evaluate how model performance changes as the amount of logged data increases. We consider sample sizes ranging from $N = 500$ to $N = 10^6$. We compare two propensity estimation methods: the existing frequency-based propensity estimation, our proposed feature-based propensity MLP, with using oracle propensities i.e., the true propensity values. We also include two reference points: the performances of the logging policy (baseline) and a ranking model trained directly on the ground-truth relevance labels (skyline).

We first evaluate whether feature-based propensity estimation improves IPS-based LTR. The results in the top row of Figure 4 show clear NDCG improvements from using the propensity MLP instead of frequency-based estimation across all datasets. For MSLR-WEB30K and Yahoo!, these gains appear across sample sizes, improving performance by approximately 2% and 1.5%, respectively. For Istella-S, the gains are strongest at smaller sample sizes and decrease as more samples are observed, which is consistent with frequency-based estimates recovering as observations increase. The propensity MLP also performs similarly to the oracle propensity baseline and even outperforms it in several settings. Although oracle propensities are unbiased with respect to the logging policy, this does not guarantee the best empirical learning signal for finite samples, as sampled data may deviate from the expected logging distribution, while the learned propensity model may provide smoother or more stable rewards during neural optimization. These findings are supported by the significance testing reported in Table 1. We find that for all sample sizes up to $N = 10^5$, feature-based propensity estimation outperforms frequency-based estimation. We find considerable and statistically significant gains across most sample sizes for MSLR-WEB30K and Yahoo!. The frequency-based propensity estimator outperforms the MLP only on Istella-S when $N = 10^6$, although these results are not statistically significant. Overall, feature-based propensity estimation strongly improves IPS performance across sample sizes, often matching or outperforming oracle propensities, with the largest and most consistent gains on MSLR-WEB30K and Yahoo! and more limited gains on Istella-S at larger sample sizes.

We continue by investigating the results for DM and DR under different propensity estimation methods shown in the middle and bottom rows of Figure 4. For both methods, we observe clear NDCG improvements when replacing frequency-based propensities with feature-based MLP propensities. These gains are largest in low-sample settings, especially between 500 and 5000 sessions, and are most pronounced on MSLR-WEB30K and Yahoo!. This suggests that feature-based propensity estimation improves the training objective for the regression model, enabling more accurate relevance

Table 1: IPS NDCG results for propensity estimation methods across sample counts. Cells show mean (SD); best non-oracle value is marked bold. Markers indicate two-sided t-test significance versus the frequency-based method: $p < 0.01$ (▲, ▼).

	Propensity	N = 10^3	N = 10^4	N = 10^5	N = 10^6
MSLR	Oracle	0.337 (0.011)	0.417 (0.026)	0.490 (0.009)	0.505 (0.003)▲
	Freq.-Based	0.335 (0.009)	0.385 (0.022)	0.478 (0.012)	0.492 (0.001)
	MLP	**0.338 (0.012)**	**0.440 (0.008)**▲	**0.496 (0.007)**▲	**0.515 (0.001)**▲
Istella-S	Oracle	0.358 (0.034)▲	0.681 (0.012)▲	0.754 (0.007)	0.791 (0.006)
	Freq.-Based	0.317 (0.034)	0.649 (0.020)	0.748 (0.017)	**0.792 (0.003)**
	MLP	**0.375 (0.038)**▲	**0.688 (0.010)**▲	**0.750 (0.004)**	0.783 (0.007)
Yahoo!	Oracle	0.720 (0.001)▲	0.767 (0.004)▲	0.786 (0.004)▲	0.792 (0.003)▲
	Freq.-Based	0.718 (0.002)	0.752 (0.010)	0.779 (0.004)	0.785 (0.001)
	MLP	**0.720 (0.001)**	**0.765 (0.005)**	**0.787 (0.005)**▲	**0.795 (0.002)**▲

Table 2: DM NDCG results for propensity estimation methods across sample counts. Cells show mean (SD); best non-oracle value is marked bold. Markers indicate two-sided t-test significance versus the frequency-based method: $p < 0.01$ (▲, ▼).

	Propensity	N = 10^3	N = 10^4	N = 10^5	N = 10^6
MSLR	Oracle	0.414 (0.019)	0.455 (0.022)▲	0.472 (0.012)	0.480 (0.013)
	Freq.-Based	0.388 (0.032)	0.440 (0.021)	0.465 (0.009)	0.470 (0.009)
	MLP	**0.430 (0.066)**	**0.481 (0.007)**▲	**0.495 (0.005)**▲	**0.504 (0.001)**▲
Istella-S	Oracle	0.669 (0.022)▲	0.736 (0.015)	0.759 (0.004)	0.775 (0.003)
	Freq.-Based	0.643 (0.025)	0.728 (0.016)	**0.758 (0.009)**	**0.772 (0.004)**
	MLP	**0.695 (0.011)**▲	**0.732 (0.008)**	0.754 (0.003)	0.772 (0.004)
Yahoo!	Oracle	0.762 (0.007)▲	0.779 (0.005)▲	0.782 (0.003)	0.782 (0.003)
	Freq.-Based	0.756 (0.008)	0.768 (0.008)	0.778 (0.003)	0.779 (0.004)
	MLP	**0.767 (0.006)**	**0.782 (0.004)**▲	**0.784 (0.004)**	**0.788 (0.002)**▲

Table 3: DR NDCG results for propensity estimation methods across sample counts. Cells show mean (SD); best non-oracle value is marked bold. Markers indicate two-sided t-test significance versus the frequency-based method: $p < 0.01$ (▲, ▼).

	Propensity	N = 10^3	N = 10^4	N = 10^5	N = 10^6
MSLR	Oracle	0.414 (0.019)	0.450 (0.021)	0.465 (0.019)	0.486 (0.009)
	Freq.-Based	0.389 (0.032)	0.437 (0.022)	0.463 (0.007)	0.493 (0.004)
	MLP	**0.448 (0.021)**	**0.473 (0.011)**▲	**0.500 (0.009)**▲	**0.512 (0.003)**▲
Istella-S	Oracle	0.670 (0.021)▲	0.735 (0.011)	0.760 (0.008)	0.781 (0.003)▼
	Freq.-Based	0.645 (0.023)	0.732 (0.009)	0.761 (0.013)	**0.790 (0.005)**
	MLP	**0.694 (0.010)**▲	**0.735 (0.007)**	**0.762 (0.008)**	0.787 (0.004)
Yahoo!	Oracle	0.762 (0.007)▲	0.777 (0.004)▲	0.776 (0.005)	0.779 (0.002)▼
	Freq.-Based	0.756 (0.008)	0.768 (0.004)	0.781 (0.003)	0.785 (0.002)
	MLP	**0.767 (0.007)**▲	**0.777 (0.004)**	**0.781 (0.002)**	**0.787 (0.002)**

estimates with fewer samples. Compared with the oracle propensity model, the propensity MLP achieves similar performance and even outperforms it in some settings. Notably, frequency-based propensity estimates outperform the oracle setting for DR on Yahoo!, although these results are within each other's prediction intervals. The significance tests in Tables 2 and 3 confirm the same pattern for both DM and DR: feature-based propensity estimation is often significantly better in lower-data settings. Differences shrink as the number of sessions increases. For MSLR-WEB30K and Yahoo!, the propensity MLP remains consistently stronger than the frequency-based estimator, gains on Istella-S are smaller. Overall, these results show that feature-based propensity estimation improves DM and DR across sample sizes and datasets, although the effect size depends on the dataset and estimator.

Table 4: Estimated position-bias results (α_k are estimated not given) for feature-based propensity estimation on $N = 10^4$ simulated queries. Cells show mean (SD) NDCG; best non-oracle value is marked bold. Markers indicate two-sided t-test significance vs frequency-based method: $p < 0.01$ (▲ and ▼).

	Propensity	IPS	DM	DR
MSLR	Oracle	0.381 (0.027)▲	0.423 (0.025)▲	0.399 (0.029)▲
	Freq.-Based	0.347 (0.014)	0.350 (0.026)	0.341 (0.029)
	MLP	**0.442 (0.009)**▲	**0.475 (0.009)**▲	**0.443 (0.027)**▲
Istella-S	Oracle	0.640 (0.033)▲	0.701 (0.024)▲	0.691 (0.028)▲
	Freq.-Based	0.448 (0.078)	0.479 (0.057)	0.502 (0.054)
	MLP	**0.685 (0.011)**▲	**0.727 (0.008)**▲	**0.722 (0.014)**▲
Yahoo!	Oracle	0.757 (0.007)▲	0.772 (0.006)▲	0.766 (0.006)▲
	Freq.-Based	0.720 (0.007)	0.731 (0.015)	0.734 (0.011)
	MLP	**0.764 (0.007)**▲	**0.777 (0.006)**▲	**0.774 (0.005)**▲

Effect of Logging Policy Randomness. In the previous section, we showed that feature-based propensity estimation improves performance across sample sizes for a logging policy that displays random rankings in 50% of cases. Since our theoretical analysis identifies logging-policy variance as a key source of bias in frequency-based propensity estimation, we now study how this effect changes with the randomness of the logging policy. To this end, we evaluate IPS, DM, and DR in the $N = 10^4$ setting across randomness levels $\epsilon \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$, reporting NDCG and comparing against oracle propensities and the frequency-based logging-policy. Results of this experiment are shown in Figure 5.

We observe that across all evaluated settings and datasets, feature-based propensities outperform the frequency-based method. For MSLR-WEB30K and Yahoo!, we observe that feature-based propensities perform better consistently. The smallest differences are found in the deterministic ($\epsilon = 0.0$) and random ($\epsilon = 1$) settings. This pattern can be explained by the fixed ranking at $\epsilon = 0.0$, where frequency-based propensities are correct from the first sample and incur no bias under Eq. 15. For Istella-S, we observe limited differences for DM and DR, possibly due to the logging policy having little effect, as the oracle propensity setting also does not improve over frequency-based estimation. Overall, we find that feature-based propensity estimation is robust across logging-policy randomness, with the strongest effects for IPS and smaller but still visible gains for DM and DR.

Effect of Estimated Position Bias. In previous sections, the propensity estimators were given the true position bias α values, as this allowed us to isolate the effect of propensity estimation methods. To further verify the effectiveness of feature-based propensity estimation, we now evaluate propensity estimation methods in settings where the position bias is also estimated. We apply the all-pairs position bias estimate, following the implementation of Hager et al. [12]. Results of the evaluation are shown in Table 4, where we show results for IPS, DM and DR for $N = 10^4$ samples. The oracle propensity in this setting means that the logging policy is the simulated placement probability, while the position bias is the estimate. We observe that across all methods and datasets, performance is significantly higher when the MLP for feature-based propensity estimation is applied. We find that the performance of feature-based

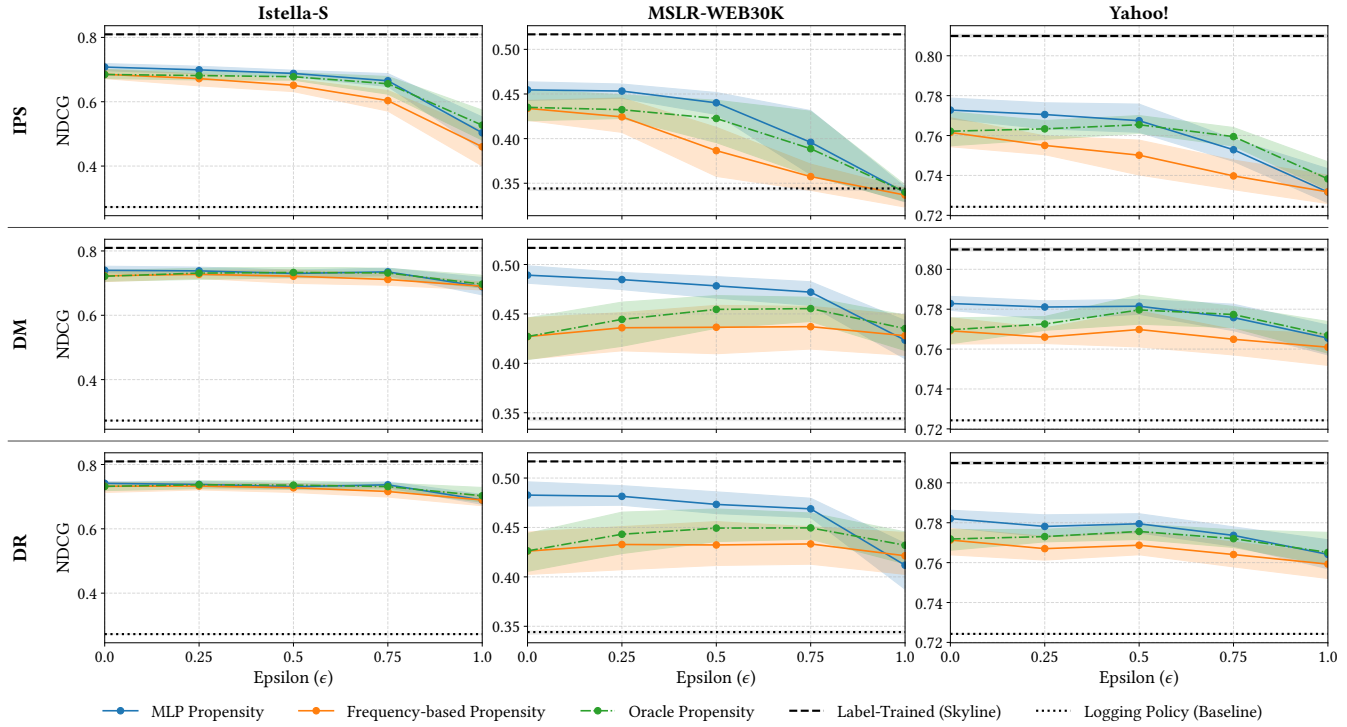


Figure 5: Performance of IPS estimation with different propensity estimation methods on NDCG in the real-target experiments at $N = 10^4$ sessions, shown for Istella-S, MSLR-WEB30K, and Yahoo! as the logging policy randomness (epsilon) increases. Results are means over 20 independent runs, shaded areas indicate the 90% prediction intervals.

propensities is also consistently higher than the performance of oracle propensity estimates. A possible explanation for why the feature-based propensity estimate may perform especially well in this setting is that, due to the smoothing that is introduced by the MSE objective, the model is less affected by variability in position bias estimates. Overall, these results verify the effectiveness of feature-based propensity estimates when position bias is estimated.

To conclude this section, we answer **RQ2** in the affirmative: feature-based propensity estimation improves counterfactual LTR performance across datasets, sample sizes, logging-policy randomness, and estimated position bias. Gains are consistent in low-data settings, but can also persist when observations are not sparse.

9 Conclusion

In this work, we introduced propensity estimation based on query-document features for counterfactual LTR. We first analyzed the theoretical and empirical problems when applying frequency-based propensity estimation methods in the often-overlooked long-tail query distribution setting. We showed that frequency-based propensity estimates result in enormous variance when fewer observations are available, directly affecting downstream off-policy evaluation and optimization. We then identified conditions under which propensity estimation can reliably use query-document features, showing that this is the case when logging policy behavior is solely based on the given features. We evaluated using MLP predicted feature-based propensities for IPS, DM, and DR estimators, in a diverse set of experimental settings. Our results reveal strong performance improvements for all models, with especially large gains for IPS-based methods. We find these improvements generalize across dataset sizes and levels of logging policy randomness, and in some cases, feature-based propensities outperformed the oracle propensity baseline, likely due to reduced variance in downstream optimization.

With feature-based propensity estimation, we hope to provide practitioners with a method for accurate and tractable propensity estimates even when individual query-document pairs have few observations, contributing to more effective and applicable counterfactual learning-to-rank in sparse and long-tail settings.

This work is the first to propose estimating propensities in a feature-based manner. We find that even in the relatively simple two proposed methods for feature-based propensity estimation, substantial improvements were found. This opens a new research direction in investigating effectively estimating propensities, for example through investigating other loss functions or clustering techniques. Other directions could verify the robustness of feature-based methods, across logging-policy sampling methods, or when logging-policy features are only partially observable (as explored in Hager et al. [13]). Finally, further work could verify the effectiveness of feature-based propensity in real-world search applications, which are often particularly affected by long-tail query distributions.

Acknowledgments

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